

# Telecom Customer Churn Analysis

## 1. Executive Summary

## 2. Business Objective

The goal of this project is to analyze customer churn in a telecom company, identify key factors associated with churn, and provide actionable recommendations to improve customer retention.

This analysis focuses on answering the following questions:

- What is the overall churn rate?
- Which customer segments are more likely to churn?
- What service, contract, or pricing factors are associated with churn?
- What retention strategies can be recommended based on the findings?

## 3. Dataset Overview

This dataset contains customer-level telecom data, with each row representing one customer.

The dataset includes information across several areas:

- Customer demographics: age, gender, marital status, dependents, city, and ZIP code
- Service information: phone service, internet service, internet type, streaming services, tech support, and security services
- Contract and billing information: contract type, payment method, paperless billing, monthly charges, total charges, refunds, and total revenue
- Churn information: customer status, churn category, and churn reason

The main outcome variable is `Customer_Status`, which indicates whether a customer stayed, churned, or recently joined.

This dataset will be used to explore churn patterns, compare customer segments, and identify potential retention opportunities.

## 4. Data Cleaning & Validation

## 4.1 Data Type Validation

Reviewed all columns to ensure correct data types for analysis.

Adjustments Made

- `Zip_Code` → changed to **Text**
- `Avg_Monthly_Long_Distance_Charges` → changed to **Decimal Number**
- `Avg_Monthly_GB_Download` → changed to **Whole Number**

## 4.2 Missing Value Handling

Checked all empty values, including:

- blank cells
- spaces
- null values

Where missing values were business-valid, they were replaced with meaningful labels or default values.

### 4.2.1 Avg\_Monthly\_Long\_Distance\_Charges

- Null values replaced with `0`

Reason: customers without long-distance usage likely incurred no charges.

### 4.2.2 Multiple\_Lines

- Blank values replaced with `No Phone Service`

### 4.2.3 Internet\_Type

- Blank values replaced with `No Internet Service`

### 4.2.4 Avg\_Monthly\_GB\_Download

- Null values replaced with `0`

### 4.2.5 Internet Service Related Columns

Blank values in the following fields were replaced with `No Internet Service`:

- Online\_Security
- Online\_Backup
- Device\_Protection\_Plan
- Premium\_Tech\_Support

- Streaming\_TV
- Streaming\_Movies
- Streaming\_Music
- Unlimited\_Data

#### 4.2.6 Churn\_Category / Churn\_Reason

Blank values replaced with `Not Churn`

Reason: active customers do not have churn reasons.

### 4.3 Outlier Review & Business Validation

Used Power Query profiling tools to inspect:

- value distribution
- minimum / maximum values
- averages
- potential anomalies

A notable anomaly was found:

**Negative values in** `Monthly_Charge`

Rather than removing these records immediately, cross-validation was performed.

#### 4.3.1 Helper Columns Created

Monthly\_Charge\_Type

- Negative Charge → `Monthly_Charge < 0`
- Positive Charge → `Monthly_Charge >= 0`

Total\_Refunds\_Type

- Refund → `Total_Refunds > 0`
- No Refund → `Total_Refunds = 0`

#### 4.3.2 Validation: Were Negative Monthly Charges Caused by Refunds?

| Charge Type     | Refund Rate |
|-----------------|-------------|
| Negative Charge | 11.67%      |
| Positive Charge | 7.38%       |

#### Finding

Customers with negative monthly charges had a higher refund rate.

**However:**

- **88.33%** of negative-charge customers had no refund record.

**Conclusion**

Negative monthly charges cannot be fully explained by refunds alone.

### 4.3.3 Validation: Are Negative-Charge Customers More Likely to Churn?

| Charge Type     | Churn Rate |
|-----------------|------------|
| Negative Charge | 25.00%     |
| Positive Charge | 26.56%     |

**Finding**

Customers with negative monthly charges were **not more likely to churn**.

### 4.3.4 Validation: Revenue Impact

| Charge Type     | Avg Refund | Avg Total Revenue |
|-----------------|------------|-------------------|
| Negative Charge | 3.18       | 2727              |
| Positive Charge | 1.94       | 3039              |

**Finding**

- Negative-charge customers had slightly higher refunds.
- Their total revenue was somewhat lower, but not materially different.

**Conclusion**

No evidence that these records significantly distort revenue metrics.

### 4.3.5 Final Assessment of Negative Monthly Charges

- Approximately **1.7%** of customers had negative monthly charges.
- Small subgroup with limited impact on overall data.
- No strong distortion to churn or revenue KPIs.
- Likely related to billing adjustments, credits, or refund processes.

**Decision**

- Retained original records
- Not treated as dirty data

## 4.4 Duplicate Check

Used `Customer_ID` to check for duplicate records.

### Result

- No duplicate customer IDs found.

## 4.5 Text Standardization

Applied:

- Trim
- Clean

To remove hidden spaces and non-printable characters in text fields.

## 4.6 Category Consistency Check

Reviewed all categorical fields.

### Result

- Category values were already standardized.

## 4.7 Data Logic Validation

Reviewed cross-column business logic.

Examples:

- No Internet Service ↔ related service fields blank / inactive
- Non-churn customers ↔ no churn reason

### Result

- No major logical inconsistencies found.

## 4.8 Final Cleaning Outcome

Dataset was successfully cleaned and validated for exploratory analysis and dashboard development.

# 5. Exploratory Data Analysis

## 5.1 Overall Churn Overview

This section provides a high-level view of the company's **current customer churn situation**, including customer loss scale, revenue exposure, and net customer growth performance.

### 5.1.1 Core KPI Summary

| KPI                            | Value    |
|--------------------------------|----------|
| Total Customers                | 7,043    |
| Churned Customers              | 1,869    |
| Churn Rate                     | 26.5%    |
| Stayed Customers               | 4,720    |
| Joined Customers               | 454      |
| Net Customer Growth            | -1,415   |
| Total Revenue                  | \$21.37M |
| Revenue from Churned Customers | \$3.68M  |
| Churn Revenue Share            | 17.2%    |

### 5.1.2 Key Findings

#### Finding 1: Customer churn is a meaningful retention challenge.

The company's churn rate is **26.5%**, meaning approximately **1 in every 4 customers has churned**.

This indicates customer retention should be treated as a key business priority.

| Status  | Share |
|---------|-------|
| Stayed  | 67.0% |
| Churned | 26.5% |
| Joined  | 6.4%  |

#### Finding 2: Customer acquisition is not offsetting churn volume.

Only **454 new customers joined**, while **1,869 customers churned**, resulting in a **net decline of 1,415 customers**.

This suggests current acquisition efforts are insufficient to maintain customer base growth.

| Status        | Count |
|---------------|-------|
| Churned       | 1,869 |
| ChurnedJoined | 454   |

#### Finding 3: Churn has material revenue impact.

Churned customers account for **\$3.68M**, representing **17.2% of total revenue**.

Although churned customers represent a larger share of customers than revenue, churn still creates meaningful revenue leakage.

| KPI                            | Value   |
|--------------------------------|---------|
| Revenue from Churned Customers | \$3.68M |
| Churn Revenue Share            | 17.2%   |

**Finding 4: Churn may be concentrated among lower-value or early-stage customers.**

Churned customers represent **26.5% of customers**, but only **17.2% of revenue**.

This suggests churn may be more common among customers with lower cumulative revenue or shorter tenure.

This requires deeper segment analysis in the next section.

| KPI                 | Value |
|---------------------|-------|
| Churn Rate          | 26.5% |
| Churn Revenue Share | 17.2% |

Churn Revenue Share < Churn Rate

↓

Monthly Charge / Tenure

↓

Contract / New Customer / Services

## 5.2 Who is Churning

### 5.2.1 Tenure Analysis: Are new customers more likely to churn?

#### Objective

To analyze whether customer tenure (length of service) impacts churn rate, and to identify high-risk lifecycle stages.

#### Methodology

- Customers were segmented into lifecycle-based tenure groups (**0–6, 6–12, 12–24, 24+ months**)
- Churn rate was calculated as: **Churned Users / Total Users**

#### Results

| Tenure Group | Total Users | Churned Users | Churn Rate |
|--------------|-------------|---------------|------------|
| 0–6 months   | 1360        | 744           | 54.7%      |
| 6–12 months  | 709         | 255           | 36.0%      |
| 12–24 months | 1047        | 309           | 29.5%      |
| 24+ months   | 3927        | 561           | 14.3%      |

## Key Findings

1. The **0–6 months group** has the highest churn rate (**54.7%**). The **24+ months group** has the lowest churn rate (**14.3%**)
2. Churn rate **decreases consistently as tenure increases**, indicating a strong negative relationship between tenure and churn
3. The gap between highest and lowest groups is **40+ percentage points**

This represents a **significant structural difference**, not random variation

4. All tenure groups contain sufficient sample sizes (**700+ users**)  
Each group includes a substantial number of churn events (**250+**), ensuring stable comparisons

## Insight

1. Churn is heavily concentrated in the **early customer lifecycle stage**
2. The first **6 months** represent a **critical risk period**
3. Customers who remain beyond this stage are significantly more likely to stay

## Business Implications

1. Churn risk is concentrated in early tenure stages
2. Retention efforts should prioritize customers within the first 6 months
3. Improving onboarding and early engagement may help reduce churn

## 5.2.2 Contract Analysis: Do contract types impact churn?

### Objective

To analyze whether contract type influences churn behavior, and identify which customer segments are at higher risk.

### Methodology

- Customers were grouped by contract type(**Month-to-month,One year,Two year**)
- Churn rate was calculated as: $\text{Churn Rate} = \frac{\text{Churned Users}}{\text{Total Users}}$

### Results

| Contract Type  | Total Users | Churned Users | Churn Rate |
|----------------|-------------|---------------|------------|
| Month-to-month | 3549        | 1625          | 45.8%      |
| One year       | 1520        | 166           | 10.9%      |
| Two year       | 1854        | 48            | 2.6%       |

## Key Findings

1. Month-to-month customers have the highest churn rate (45.8%), while two-year contract customers have the lowest (2.6%)
2. Churn rate decreases significantly as contract length increases, showing a strong negative relationship between contract commitment and churn
3. The gap between short-term and long-term contracts exceeds 40 percentage points, indicating a substantial and actionable difference

### Insight

1. Customers without long-term commitment are significantly more likely to churn
2. Longer contracts act as a strong retention mechanism
3. Contract type is one of the strongest predictors of customer stability

### Business Implications

1. Customers on month-to-month contracts should be prioritized for retention efforts
2. Encouraging migration to longer-term contracts may reduce churn
3. Contract type should be incorporated into retention targeting strategies

## 5.2.3 Pricing Analysis: Do monthly charge impact churn?

### Objective

To evaluate whether monthly pricing level is associated with customer churn, and determine if pricing is a relevant factor for further analysis.

### Methodology

- Customers with non-negative monthly charges were segmented into three pricing tiers using **quantile-based segmentation (NTILE): Low (bottom 33%), Medium (middle 33%), High (top 33%)**

This approach ensures **balanced group sizes**, allowing for more reliable comparison across segments

- Churn rate was calculated as: **Churn Rate = Churned Users / Total Users**

### Result

| Charge Group | Total Users | Churned Users | Churn Rate |
|--------------|-------------|---------------|------------|
| Low          | 2308        | 360           | 16%        |
| Medium       | 2308        | 688           | 30%        |
| High         | 2307        | 791           | 34%        |

### Key Findings

1. Churn rate increases consistently across pricing tiers

2. The difference between Low and High segments is approximately **18 percentage points**, indicating a substantial gap
3. Balanced group sizes (~2300 users per group) improve the reliability of the comparison and reduce sampling bias

### Insight

1. Monthly charge shows a strong positive association with churn, as customers in higher pricing tiers exhibit significantly higher churn rates.
2. However, this analysis does not control for key structural variables such as contract type and customer tenure. Therefore, it is unclear whether pricing is a **primary driver of churn** or a **conditional factor influenced by customer segments**.
3. Further segmentation is required to validate its impact.

### Business Implications

1. Pricing may play a role in customer churn and should be considered in further analysis
2. Additional segmentation is required to identify which customer groups are most sensitive to pricing

## 5.3 Where is Churn Concentrated? Cross-Segement Analysis

### 5.3.1 Cross Analysis: Tenure × Contract

#### Objective

To identify high-risk customer segments by analyzing the combined impact of **customer tenure** and **contract type** on churn.

#### Methodology

- Customers were segmented into lifecycle-based tenure groups (**0–6, 6–12, 12–24, 24+ months**) and contract type groups (**Month-to-month, One year, Two year**)
- Churn rate was calculated as: **Churned Users / Total Users**

#### Results

| Contract       | Tenure       | Total Users | Churned Users | Churn Rate |
|----------------|--------------|-------------|---------------|------------|
| Month-to-month | 0–6 months   | 1276        | 740           | 58%        |
| Month-to-month | 6–12 months  | 553         | 251           | 45%        |
| Month-to-month | 12–24 months | 718         | 288           | 40%        |
| Month-to-month | 24+ months   | 1063        | 376           | 35%        |
| One-year       | 0–6 months   | 44          | 4             | 9%         |
| One-year       | 6–12 months  | 80          | 4             | 5%         |

| Contract | Tenure       | Total Users | Churned Users | Churn Rate |
|----------|--------------|-------------|---------------|------------|
| One-year | 12–24 months | 216         | 21            | 10%        |
| One-year | 24+ months   | 1210        | 137           | 11%        |
| Two-year | 0–6 months   | 40          | 0             | 0%         |
| Two-year | 6–12 months  | 76          | 0             | 0%         |
| Two-year | 12–24 months | 113         | 0             | 0%         |
| Two-year | 24+ months   | 1654        | 48            | 3%         |

## Key Findings

### 1. Contract type is the strongest determinant of churn

- Month-to-month customers consistently show the highest churn rates across all tenure groups
- Two-year contract customers exhibit extremely low churn across all lifecycle stages

This indicates that **contract structure plays a dominant role in retention**

### 2. Tenure impact varies significantly by contract type

- For Month-to-Month users:
  - Churn decreases steadily with tenure (58% → 35%)
  - Indicates a strong lifecycle effect
- For One-Year users:
  - Churn remains relatively stable (~5%–11%)
  - No clear downward trend
- For Two-Year users:
  - Churn is consistently low across all stages

### 3. Early-stage churn is concentrated in low-commitment users

The highest churn (58%) occurs in:

- 0–6 month + Month-to-Month segment

In contrast:

- Early-stage users under long-term contracts show minimal churn

Indicates that **lack of commitment amplifies early-stage churn risk**

## Insight

1. Churn is primarily driven by **contract commitment**, with tenure acting as a secondary, segment-dependent factor

2. The effect of tenure is **strongest among flexible (Month-to-Month) users**, but becomes negligible under long-term contracts
3. High churn risk is highly concentrated in **early-stage, low-commitment customers**, rather than being uniformly distributed across the lifecycle

### Business Implications

1. Retention efforts should prioritize early-stage month-to-month customers
2. Encouraging migration to long-term contracts could significantly reduce churn
3. Early lifecycle engagement remains important, especially for non-committed users

## 5.3.2 Cross Analysis: Monthly Charge × Contract

### Objective

To examine how the impact of monthly pricing on churn varies across different contract types, and determine whether pricing sensitivity is consistent or moderated by customer commitment level.

### Methodology

- Customers with non-negative monthly charges were segmented into three pricing tiers using **quantile-based segmentation (NTILE): Low (bottom 33%), Medium (middle 33%), High (top 33%)**
- Cross-analysis was conducted with **Contract Type** (Month-to-Month, One Year, Two Year)
- Churn rate was calculated as: **Churn Rate = Churned Users / Total Users**
- The analysis focuses on: Price impact **within each contract group**, differences in **price sensitivity across contract types**

### Result

| Contract       | Charge Group | Total Users | Churn Rate |
|----------------|--------------|-------------|------------|
| Month-to-Month | Low          | 1102        | 30%        |
| Month-to-Month | Medium       | 1324        | 48%        |
| Month-to-Month | High         | 1123        | 58%        |
| One Year       | Low          | 479         | 4%         |
| One Year       | Medium       | 498         | 9%         |
| One Year       | High         | 543         | 19%        |
| Two Year       | Low          | 727         | 1%         |
| Two Year       | Medium       | 486         | 2%         |
| Two Year       | High         | 641         | 5%         |

## Key Findings

### 1. Churn rate increases consistently with pricing across all contract types

- Higher pricing tiers are associated with higher churn in Month-to-Month, One Year, and Two Year contracts
- This confirms that pricing has a **directionally consistent effect**

### 2. Price sensitivity varies significantly by contract type

- Churn gap (High vs Low):
  - Month-to-Month: **28 percentage points**
  - One Year: **15 percentage points**
  - Two Year: **4 percentage points**
- The magnitude of the gap decreases as contract length increases

### 3. Contract type acts as a moderating variable

- While pricing impacts churn across all segments, the **strength of the relationship weakens with longer contracts**
- Long-term contracts significantly reduce overall churn levels

## Insight

1. Monthly charge is not only associated with churn, but its impact is **conditional on customer commitment level**
2. Pricing has the strongest effect on customers with **low commitment (Month-to-Month)**, indicating high price sensitivity in flexible segments
3. For customers under long-term contracts, churn remains low even at higher pricing levels, suggesting that:
  - Contractual commitment reduces responsiveness to pricing
  - Other factors (e.g., service quality, experience) may play a more dominant role

## Business Implications

1. **Different pricing strategies are needed for different customer segments**
2. **Contract upgrades can mitigate price-driven churn**
3. **Pricing alone is not sufficient for retention in all segments**

## 5.3.3 Cross Analysis: Monthly Charge × Tenure

### Objective

To evaluate how the relationship between monthly pricing and churn varies across different lifecycle stages, and determine whether price sensitivity is consistent or moderated by customer tenure.

## Methodology

- Customers with non-negative monthly charges were segmented into three pricing tiers using **quantile-based segmentation (NTILE): Low (bottom 33%), Medium (middle 33%), High (top 33%)**
- Customers were grouped into lifecycle-based tenure segments: **0–6 months, 6–12 months, 12–24 months, 24+ months**
- Cross-analysis was conducted with **Tenure Group**
- Churn rate was calculated as: **Churn Rate = Churned Users / Total Users**

## Results

| Tenure Group | Charge Group | Total Users | Churned Users | Churn Rate |
|--------------|--------------|-------------|---------------|------------|
| 0–6 months   | High         | 187         | 146           | 78.1%      |
| 0–6 months   | Medium       | 543         | 349           | 64.3%      |
| 0–6 months   | Low          | 604         | 232           | 38.4%      |
| 6–12 months  | High         | 155         | 112           | 72.3%      |
| 6–12 months  | Medium       | 267         | 96            | 36.0%      |
| 6–12 months  | Low          | 272         | 44            | 16.2%      |
| 12–24 months | High         | 290         | 154           | 53.1%      |
| 12–24 months | Medium       | 357         | 109           | 30.5%      |
| 12–24 months | Low          | 377         | 40            | 10.6%      |
| 24+ months   | High         | 1675        | 379           | 22.6%      |
| 24+ months   | Medium       | 1141        | 134           | 11.7%      |
| 24+ months   | Low          | 1055        | 44            | 4.2%       |

## Key Findings

1. Higher pricing tiers are consistently associated with higher churn across all lifecycle stages  
Within each tenure group, customers in higher pricing tiers generally exhibit higher churn rates  
This suggests that pricing is associated with churn behavior across the customer lifecycle  
However, earlier cross-analysis showed that pricing effects are not independent, and may be influenced by structural factors such as contract commitment.

2. Churn decreases steadily as tenure increases across all pricing tiers  
For all pricing levels, churn rate declines significantly as tenure increases

Example (High-price group):

0–6 months: 78.1%

24+ months: 22.6%

This indicates a strong lifecycle stabilization effect, where long-tenure customers become substantially less likely to churn.

### 3. Price sensitivity appears strongest during the early lifecycle stage

The churn gap between High and Low pricing tiers:

- 0–6 months: ~40%
- 6–12 months: ~56%
- 12–24 months: ~42%
- 24+ months: ~18%

The pricing gap narrows significantly among long-tenure customers.

This suggests that customers may become less responsive to pricing as customer relationships mature.

### 4. Churn risk is highly concentrated among early-stage, high-price customers

Highest churn segment:

- 0–6 months + High price → 78.1%

Lowest churn segment:

- 24+ months + Low price → 4.2%

This indicates strong concentration of churn risk within specific lifecycle and pricing combinations.

## Insight

1. Pricing is associated with higher churn, but its impact appears to be moderated by customer lifecycle stage.
2. Customers in the early lifecycle stage exhibit substantially higher responsiveness to pricing exposure.
3. As tenure increases, churn decreases significantly across all pricing groups, suggesting stronger customer stability over time.
4. Earlier cross-analysis also showed that pricing effects become significantly stronger among low-commitment (month-to-month) customers, indicating that customer commitment structure plays an important moderating role.

5. Churn risk is highly concentrated among early-stage, high-price, low-commitment customers.

### **Business Implications**

1. Retention efforts should prioritize early-stage, high-price customers, particularly those under low-commitment contract structures.
2. Lifecycle-based retention and pricing strategies may be more effective than uniform pricing policies.
3. Increasing customer commitment may help reduce pricing-related churn sensitivity.
4. Pricing optimization alone may not fully address churn risk without improving customer retention structure and long-term engagement.

### **5.3.4 Cross-Segment Summary**

Cross-analysis suggests that customer churn is not driven equally across all customer groups.

The findings indicate that:

- Contract type is the dominant structural driver of churn
- The effects of tenure and pricing are highly conditional on customer commitment level
- Early-stage and high-price churn risks become significantly amplified among month-to-month customers

This suggests that churn risk is primarily concentrated among low-commitment customers who are more vulnerable during the early lifecycle stage and more sensitive to pricing exposure.

## **5.4 Why Are Customers Churning? Root Cause Analysis**

### **5.4.1 High-Risk Segment Definition**

High-risk segment = early-stage + month-to-month + high-price

### **5.4.2 Churn Category Analysis: High-Risk Segment vs Other Customers**

#### **Objective**

To compare churn category distribution between the identified high-risk segment and other customer groups, and determine whether high-risk customers exhibit distinct churn patterns or primarily demonstrate greater structural churn vulnerability.

#### **Methodology**

- Only churned customers were included in the analysis
- Customers were divided into:
  - High-Risk Segment

- Other Customers
- Churn categories were compared between the two groups
- Percentage contribution was calculated within each customer segment

## Results

| Customer Segment  | Churn Category  | Churned Customers | Percentage |
|-------------------|-----------------|-------------------|------------|
| High-Risk Segment | Competitor      | 71                | 49%        |
| High-Risk Segment | Dissatisfaction | 21                | 14%        |
| High-Risk Segment | Attitude        | 21                | 14%        |
| High-Risk Segment | Other           | 20                | 14%        |
| High-Risk Segment | Price           | 12                | 8%         |
| Other Customers   | Competitor      | 760               | 45%        |
| Other Customers   | Dissatisfaction | 291               | 17%        |
| Other Customers   | Attitude        | 286               | 17%        |
| Other Customers   | Price           | 197               | 12%        |
| Other Customers   | Other           | 160               | 9%         |

## Key Findings

1. Competitor-related churn is the dominant category across all customer groups

Competitor-related churn represents the largest churn category for both customer groups:

- High-Risk Segment: 49%
- Other Customers: 45%

This indicates that competitive pressure is a broad churn driver affecting the overall customer base, rather than an issue isolated to high-risk customers.

2. High-risk customers do not exhibit dramatically different churn category structures

The overall churn category distribution between high-risk customers and other customers remains relatively similar across major categories:

- Competitor
- Dissatisfaction
- Attitude
- Price

Although competitor-related churn is slightly higher within the high-risk segment, no category shows a dramatic divergence between the two groups.

This suggests that high-risk customers are not leaving for fundamentally different reasons.

3. Elevated churn among high-risk customers appears more related to structural vulnerability than unique churn triggers

Earlier cross-analysis identified that churn risk is highly concentrated among:

- early-stage customers
- month-to-month contract users
- high-price customers

However, the current comparison shows that high-risk customers do not display substantially different churn category patterns compared to the broader customer base.

This suggests that their elevated churn risk is more likely explained by structural characteristics such as:

- lower contractual commitment
- lower switching barriers
- higher sensitivity to pricing and competitive alternatives during the early customer lifecycle

rather than by a unique churn cause.

### **Insight**

1. Competitor-related churn exists broadly across the customer base and appears to be a market-wide retention challenge.
2. High-risk customers appear more susceptible to common churn triggers due to structural risk factors identified in earlier analysis.
3. The findings suggest that high churn risk may be driven more by customer vulnerability structure than by distinct churn reasons unique to the high-risk segment.

### **Business Implications**

1. Retention strategies should focus not only on churn triggers, but also on reducing structural churn vulnerability.
2. Improving customer commitment and early-stage engagement may help reduce responsiveness to competitor-related churn factors.
3. Competitor-related churn should be treated as a broad market challenge rather than a problem isolated to a single customer segment.
4. Further analysis at the detailed churn reason level is required to determine whether specific competitive factors (e.g., pricing, offers, service quality) disproportionately affect high-risk customers.

## **5.4.3 Churn Reason Analysis: High-Risk Segment vs Other Customers**

## Objective

To compare detailed churn reasons between the identified high-risk segment and other customer groups, and determine whether high-risk customers are associated with distinct churn triggers or demonstrate higher sensitivity to common churn factors.

## Methodology

- Only churned customers were included in the analysis
- Customers were divided into:
  - High-Risk Segment
  - Other Customers
- Detailed **Churn\_Reason** values were analyzed and compared between the two groups
- Percentage contribution was calculated within each customer segment
- The analysis focused on identifying:
  - differences in churn reason concentration
  - recurring competitive factors
  - service-related vulnerabilities
  - potential structural sensitivity patterns

## Results

| Customer Segment  | Churn Reason                  | Churned Customers | Percentage |
|-------------------|-------------------------------|-------------------|------------|
| High-Risk Segment | Competitor had better devices | 30                | 20.69%     |
| High-Risk Segment | Competitor made better offer  | 26                | 17.93%     |
| High-Risk Segment | Don't know                    | 14                | 9.66%      |
| High-Risk Segment | Attitude of service provider  | 13                | 8.97%      |
| High-Risk Segment | Network reliability           | 10                | 6.90%      |
| Other Customers   | Competitor had better devices | 281               | 16.59%     |
| Other Customers   | Competitor made better offer  | 278               | 16.41%     |
| Other Customers   | Attitude of support person    | 207               | 12.22%     |
| Other Customers   | Don't know                    | 114               | 6.73%      |
| Other Customers   | Competitor offered more data  | 109               | 6.43%      |

## Key Findings

1. Competitor-related reasons remain the dominant churn trigger across both customer groups

For both High-Risk Segment and Other Customers, the most common churn reasons are:

- Competitor had better devices
- Competitor made better offer

This further confirms that competitive pressure is a major driver of churn across the overall customer base.

## 2. High-risk customers appear more sensitive to perceived competitive value differences

Compared to other customers, the High-Risk Segment shows higher concentration in:

- Competitor had better devices
  - 20.69% vs 16.59%
- Competitor made better offer
  - 17.93% vs 16.41%

This suggests that high-risk customers may be more responsive to perceived value advantages offered by competitors, particularly during the early customer lifecycle stage.

## 3. Early-stage service experience appears more influential among high-risk customers

The High-Risk Segment also shows relatively higher concentration in:

- Attitude of service provider
  - 8.97% vs 4.66%
- Network reliability
  - 6.90% vs 3.60%

This suggests that customers with low commitment and short tenure may be more vulnerable to negative early service experiences.

## 4. Other customers show relatively stronger support-related churn patterns

Other Customers exhibit higher concentration in:

- Attitude of support person
  - 12.22% vs 5.52%

This may indicate that more established customers are more likely to churn after accumulated support or service interaction dissatisfaction.

## Insight

1. High-risk customers do not appear to churn due to entirely unique reasons. Instead, they appear more sensitive to common competitive and service-related churn triggers.
2. Earlier structural analysis identified that churn risk is concentrated among:
  - early-stage users

- month-to-month customers
  - high-price customers
3. The current analysis suggests that these structural characteristics increase customer responsiveness to:
- competitor value propositions
  - attractive offers and devices
  - early negative service experiences
4. The findings support the hypothesis that high churn risk is driven primarily by structural vulnerability rather than isolated churn causes.

### **Business Implications**

1. Retention strategies for high-risk customers should focus on strengthening perceived customer value during the early lifecycle stage.
2. Early-stage customer experience management may play an important role in reducing churn vulnerability.
3. Competitive positioning — particularly device offerings and promotional value — may be critical for retaining low-commitment customers.
4. Long-term customers may require different retention strategies focused more on support quality and relationship management.

## **5.5 Business Impact Analysis**

### **5.5.1 Contract Impact Analysis**

#### **Objective**

To evaluate the business impact of churn across different contract types by comparing:

- churn risk
- customer volume
- revenue loss
- customer value

The analysis aims to determine which customer segments contribute most significantly to overall business impact and should therefore be prioritized for retention efforts.

#### **Methodology**

Customers were grouped by Contract type:

- Month-to-Month

- One Year
- Two Year

For each contract group, the following metrics were calculated:

- Total Customers
- Churned Customers
- Churn Rate
- Churned Revenue
- Average Revenue per Churned Customer

The analysis evaluates both:

- churn risk
- business impact

to determine operational retention priorities.

## Results

| Contract       | Total Customers | Churned Customers | Churn Rate | Churned Revenue | Avg Revenue per Churned Customer |
|----------------|-----------------|-------------------|------------|-----------------|----------------------------------|
| Month-to-Month | 3549            | 1625              | 45.8%      | 2,459,014       | 1,513                            |
| One Year       | 1520            | 166               | 10.9%      | 858,490         | 5,172                            |
| Two Year       | 1854            | 48                | 0.026%     | 335,864         | 6,997                            |

## Key Findings

1. Month-to-month customers represent the highest-risk segment

Month-to-month customers exhibit a churn rate of approximately **45.8%**, significantly higher than both one-year and two-year contract customers.

This further confirms earlier findings that low contractual commitment is a major structural driver of churn risk.

2. Month-to-month customers also contribute the largest overall revenue loss

Despite having lower average customer value, month-to-month customers generate the highest total churned revenue:

- approximately **\$2.45M** in churned revenue
- substantially higher than all other contract groups

This indicates that the business impact of churn is heavily concentrated within the month-to-month customer base due to:

- large customer volume
- extremely high churn frequency

3. Long-term contract customers demonstrate significantly higher customer value

Although long-term contract customers exhibit much lower churn risk, their average revenue per churned customer is substantially higher:

- One Year: approximately **\$5,172**
- Two Year: approximately **\$6,997**

compared to:

- Month-to-Month: approximately **\$1,513**

This suggests that long-term customers represent a high-value segment with stronger revenue contribution on a per-customer basis.

4. High-risk customers are not always the highest-value customers

The analysis highlights an important business tradeoff:

- Month-to-month customers represent the largest operational churn problem
- Long-term contract customers represent higher individual customer value

This suggests that retention strategies may require different operational approaches depending on customer segment characteristics.

### **Insight**

1. Churn impact is not determined solely by churn risk.

2. Month-to-month customers create the largest overall business impact because of:

- high churn frequency
- large segment size
- substantial accumulated revenue leakage

3. Long-term contract customers, while relatively stable, still represent strategically important high-value customers.

4. These findings suggest that retention prioritization should balance:

- churn risk
- business impact
- customer value

- operational scalability

### **Business Implications**

1. Month-to-month customers should be prioritized for large-scale retention initiatives

Given their high churn rate and large contribution to total revenue loss, month-to-month customers represent the most operationally critical retention segment.

2. Long-term customers may require high-value retention strategies

Although long-term customers exhibit lower churn risk, their high individual customer value suggests that targeted relationship management strategies may still be important.

3. Retention strategy should differentiate between operational scale and customer value

The findings indicate that effective retention management may require:

- broad operational retention strategies for high-volume churn segments
- targeted high-value retention strategies for premium customer groups

## **5.6 Operational Segment Trigger Analysis**

### **5.6.1 Churn Reason Analysis by Contract Type**

#### **Objective**

To identify whether different operational customer segments exhibit different churn triggers by comparing churn reason distributions across contract types.

The analysis aims to determine:

- whether churn behavior differs across commitment levels
- which churn triggers are most relevant for each operational segment
- how retention strategies should be differentiated across customer groups

#### **Methodology**

Only churned customers were included in the analysis.

Customers were grouped by contract type:

- Month-to-Month
- One Year
- Two Year

For each contract group:

- churn reasons were aggregated
- percentage contribution was calculated within each contract segment

This analysis focuses on identifying operational churn triggers for high-impact customer segments.

## Results

### Month-to-Month Customers

| Churn Reason                  | Percentage |
|-------------------------------|------------|
| Competitor had better devices | 17%        |
| Competitor made better offer  | 17%        |
| Attitude of support person    | 12%        |
| Don't know                    | 7%         |
| Competitor offered more data  | 6%         |

### One-Year Customers

| Churn Reason                              | Percentage |
|-------------------------------------------|------------|
| Competitor made better offer              | 14%        |
| Competitor had better devices             | 11%        |
| Competitor offered higher download speeds | 9%         |
| Attitude of support person                | 8%         |
| Attitude of service provider              | 8%         |

### Two-Year Customers

| Churn Reason                              | Percentage |
|-------------------------------------------|------------|
| Competitor made better offer              | 17%        |
| Product dissatisfaction                   | 13%        |
| Competitor had better devices             | 13%        |
| Competitor offered more data              | 13%        |
| Competitor offered higher download speeds | 8%         |

## Key Findings

### 1. Competitor-related reasons dominate churn across all contract types

Across all contract groups, the most common churn reasons are competitor-related factors, including:

- competitor made better offer
- competitor had better devices
- competitor offered more data
- competitor offered higher download speeds

This suggests that competitive pressure is a major market-wide churn driver affecting the overall customer base.

## 2. Month-to-month customers exhibit the strongest sensitivity to competitor value propositions

Month-to-month customers show particularly high concentration in:

- competitor made better offer
- competitor had better devices
- competitor offered more data

In addition, service-related factors also appear prominently:

- attitude of support person
- attitude of service provider
- service dissatisfaction

This suggests that month-to-month customers are highly:

- value sensitive
- price sensitive
- experience sensitive

Because these customers have low contractual commitment and low switching barriers, they appear more responsive to:

- competitor incentives
- short-term service dissatisfaction
- perceived value gaps

## 3. One-year customers demonstrate greater stability and lower service sensitivity

Although competitor-related reasons remain important among one-year customers, service-related churn reasons appear less dominant compared to month-to-month customers.

This suggests that moderate contractual commitment may reduce responsiveness to short-term dissatisfaction and improve customer stability.

## 4. Long-term contract customers exhibit signs of deeper product-level dissatisfaction

Among two-year customers, churn appears less driven by short-term switching behavior and more associated with:

- product dissatisfaction
- long-term value perception
- accumulated service experience

Although churn volume is relatively small within this segment, the findings suggest that long-term customers may require different retention approaches focused on:

- relationship management
- product quality
- long-term customer satisfaction

### **Insight**

1. Churn triggers vary across operational customer segments.
2. Lower-commitment customers are more responsive to:
  - competitor offers
  - pricing/value comparisons
  - short-term service experiences
3. Higher-commitment customers appear more influenced by:
  - long-term product satisfaction
  - accumulated service quality
  - overall relationship value
4. These findings suggest that churn behavior is influenced not only by customer characteristics, but also by customer commitment level and switching barriers.

### **Business Implications**

1. Retention strategies for month-to-month customers should prioritize value competitiveness

Potential strategies include:

- retention offers
- contract upgrade incentives
- promotional bundles
- value communication campaigns

Because these customers demonstrate high responsiveness to competitive alternatives.

2. Early customer experience management is critical for low-commitment customers

Month-to-month customers also appear highly sensitive to:

- support interactions
- service experience
- onboarding quality

Potential strategies include:

- onboarding optimization
- faster issue resolution
- proactive support engagement
- customer education

3. Long-term customer retention should focus on relationship quality and product satisfaction

For long-term customers, retention strategies may require:

- premium support
- loyalty management
- proactive customer recovery
- product quality improvement
- personalized retention engagement

4. Retention strategy should be differentiated by operational segment

The findings suggest that a single retention strategy may not be equally effective across all customer groups.

Instead, retention programs should align with:

- customer commitment level
- churn trigger type
- customer value
- operational impact

## **5.7 Retention Prioritization & Strategy Recommendations**

### **5.7.1 Retention Prioritization**

#### **Objective**

To identify which customer segments should be prioritized for retention efforts based on:

- churn risk
- business impact
- customer value
- operational preventability

The goal is to determine how retention resources should be allocated across different operational customer groups.

### Retention Prioritization Framework

The prioritization framework evaluates customer segments using four dimensions:

| Dimension      | Description                                        |
|----------------|----------------------------------------------------|
| Risk           | How likely customers are to churn                  |
| Impact         | Business damage created by churn                   |
| Customer Value | Revenue contribution and long-term value           |
| Preventability | Likelihood that retention actions can reduce churn |

### Priority Segments

| Segment                   | Risk      | Impact     | Preventability | Strategic Priority |
|---------------------------|-----------|------------|----------------|--------------------|
| Month-to-Month Customers  | Very High | Very High  | High           | Highest Priority   |
| Early Lifecycle Customers | High      | High       | High           | High Priority      |
| Long-Term Customers       | Low       | High Value | Medium         | Strategic Priority |

### Key Findings

1. Month-to-month customers represent the highest operational retention priority

Month-to-month customers demonstrate:

- the highest churn risk
- the largest churn volume
- the greatest total revenue leakage

In addition, churn triggers within this segment appear highly actionable, including:

- competitor offers
- pricing/value sensitivity
- onboarding and service experience

This suggests that targeted retention initiatives within this segment may produce the largest operational impact.

2. Early lifecycle customers represent a critical stabilization stage

Earlier analysis showed that churn is heavily concentrated within the first 12 months, particularly among:

- low-commitment customers
- high-price users

This suggests that early-stage customer stabilization may significantly reduce future churn risk.

Because loyalty and switching barriers are still weak during the early lifecycle stage, customer experience management appears especially important.

3. Long-term customers represent strategically important high-value accounts

Although long-term customers exhibit relatively low churn risk, their average customer value is substantially higher.

This suggests that losing even a small number of long-term customers may still create meaningful business impact on a per-customer basis.

As a result, long-term customer retention should focus more on:

- relationship quality
- product satisfaction
- proactive service management

rather than large-scale discount-driven retention campaigns.

5.7.2 Large-Scale Retention Strategy for Month-to-Month Customers

Recommended Focus

- Reduce responsiveness to competitor offers
- Improve perceived value
- Strengthen early customer commitment
- Increase switching barriers

Potential Strategies

| Area                  | Recommended Action                                 |
|-----------------------|----------------------------------------------------|
| Pricing               | Retention discounts and bundle optimization        |
| Commitment            | Contract upgrade incentives                        |
| Customer Experience   | Faster support response and onboarding improvement |
| Value Communication   | Highlight service quality and bundled benefits     |
| Competitive Retention | Personalized retention offers for at-risk users    |

Strategic Rationale

Month-to-month customers exhibit:

- high churn probability
- strong price sensitivity

- high responsiveness to competitor value propositions
- low switching barriers

Because this segment also contributes the largest overall churn volume and revenue leakage, broad operational retention strategies may generate significant business impact.

## 5.7.3 Early Lifecycle Stabilization Strategy

### Recommended Focus

- Improve onboarding quality
- Accelerate customer value realization
- Reduce early dissatisfaction
- Increase early customer engagement

### Potential Strategies

| Area                 | Recommended Action                         |
|----------------------|--------------------------------------------|
| Onboarding           | Guided onboarding and setup support        |
| Education            | Feature education and service tutorials    |
| Engagement           | Early follow-up communication              |
| Service Recovery     | Rapid issue resolution during first months |
| Retention Monitoring | Early-risk detection system                |

### Strategic Rationale

Earlier analysis showed that churn risk is highly concentrated during the early lifecycle stage, particularly among:

- low-commitment customers
- high-price users

This suggests that improving early-stage customer experience may significantly reduce future churn probability before customer dissatisfaction and switching behavior become established.

## 5.7.4 High-Value Customer Retention Strategy

### Recommended Focus

- Maintain long-term customer satisfaction
- Protect high-value accounts
- Improve relationship quality
- Prevent accumulated dissatisfaction

## Potential Strategies

| Area                    | Recommended Action                               |
|-------------------------|--------------------------------------------------|
| Relationship Management | Dedicated support and proactive engagement       |
| Loyalty Programs        | Long-term customer rewards                       |
| Service Quality         | Premium support experience                       |
| Product Satisfaction    | Continuous service quality improvement           |
| Retention Recovery      | Personalized recovery for dissatisfied customers |

## Strategic Rationale

Although long-term customers exhibit relatively low churn risk, they contribute substantially higher revenue on a per-customer basis.

This suggests that retention strategies for long-term customers should prioritize:

- relationship quality
- product satisfaction
- customer trust

rather than short-term pricing incentives alone.

## 5.7.5 Strategic Recommendation Summary

The analysis suggests that effective churn reduction requires differentiated retention strategies across operational customer segments.

Key findings indicate that:

- low-commitment customers require large-scale operational retention programs
- early-stage customers require onboarding and stabilization support
- high-value long-term customers require relationship-focused retention strategies

Rather than applying a single retention strategy across the entire customer base, retention efforts should align with:

- customer commitment level
- churn trigger type
- business impact
- operational scalability

This approach may improve both:

- retention efficiency
- long-term customer value protection

## **6. Key Findings**

### **6.1. Customer Churn Is Primarily Concentrated Among Low-Commitment Customers**

Month-to-month customers exhibit significantly higher churn rates across nearly all analyses, indicating that low contractual commitment is the strongest structural driver of churn.

Cross-analysis further confirms that contract structure amplifies other churn factors such as pricing sensitivity and early lifecycle instability.

### **6.2. Early Lifecycle Represents the Highest Churn Vulnerability Stage**

Customers within the first 6 months show substantially higher churn risk compared to long-tenure customers.

The combination of early tenure and low commitment creates the highest-risk customer segment, suggesting that onboarding and early engagement are critical retention opportunities.

### **6.3 Pricing Sensitivity Is Strongest Among Flexible Customers**

Higher monthly charges are consistently associated with higher churn rates.

However, cross-analysis reveals that pricing impact is significantly stronger among month-to-month customers, while long-term contracts greatly reduce price sensitivity.

This suggests that pricing alone is not the root cause of churn, but rather a conditional risk factor moderated by customer commitment level.

### **6.4 Churn Creates Significant Revenue Leakage Despite Lower Customer Value**

Although churned customers account for only 17.2% of total revenue, overall churn still generates approximately \$3.68M in revenue loss.

Month-to-month customers contribute the largest portion of churned revenue, making them the highest operational retention priority.

### **6.5 Long-Term Customers Represent Strategic Retention Value**

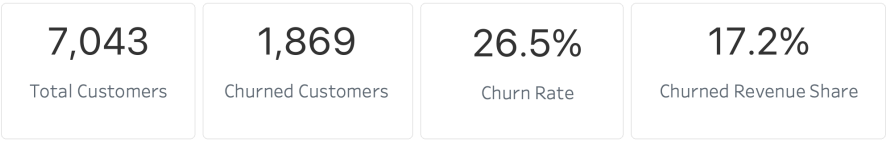
Customers under long-term contracts demonstrate extremely low churn rates while generating the highest average revenue per churned customer.

Although they are not the primary churn-risk segment, retaining these customers remains strategically important due to their long-term business value.

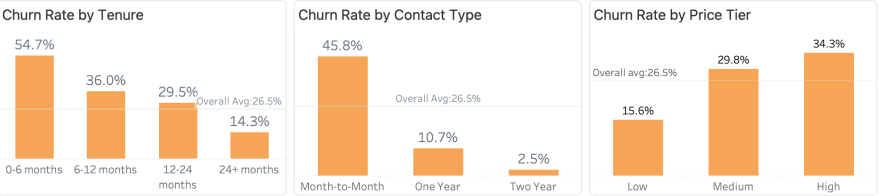
# 7. Dashboard

## Customer Churn & Retention Insights

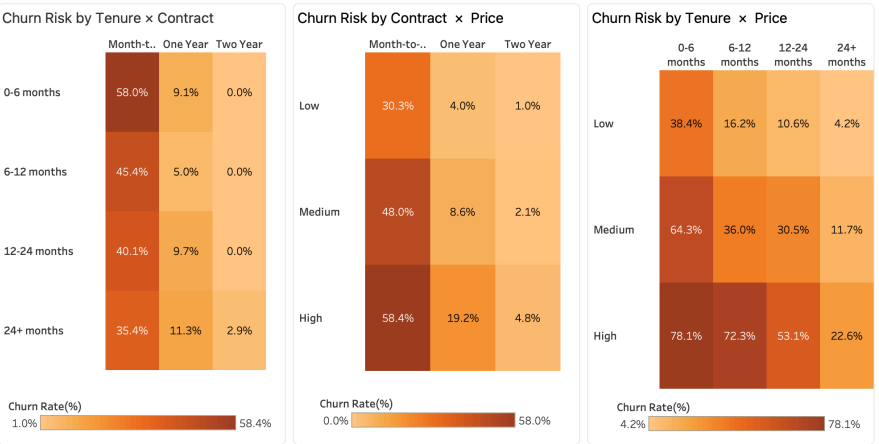
Identify high-risk segments and retention opportunities.



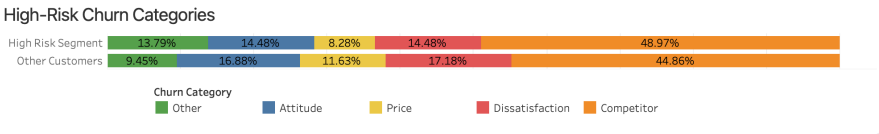
### Churn Overview



### Cross Analysis



### Diagnostic Analysis



 **Highest Risk Segment**

 **Profile**

- Tenure: 0-6 months
- Contract: Month-to-Month
- Monthly Charge: Medium to High(\$30+)
- Churn Risk: **Highest(58%)**

 **Key Findings**

- Contract type is the strongest structural driver.
- Tenure and pricing effects become significantly stronger under Month-to-Month contracts.
- Long-term contracts substantially reduce churn risk, even among high-price customers.
- High-risk customers are more vulnerable to...

 **Recommended Actions**

- Prioritize retention efforts during the first 6 months of the customer lifecycle.
- Encourage Month-to-Month customers to upgrade to long-term contracts.
- Provide targeted retention offers for high-price customers vulnerable to competitor promotions.
- Strengthen onboarding and customer engagement for early-stage users.

 **Expected Impact**

- Reduce early-stage churn risk
- Improve customer retention stability
- Lower churn-related revenue loss
- Increase customer lifetime value (CLV)

# Retention Prioritization & Strategy

Prioritize retention efforts based on churn risk, customer value and business impact.

